Budget (this was in the Reagan years) to exaggerate the uncertainties and minimize the dangers. His calculation on the explosion of the Philippine volcano Mt. Pinatubo and his prediction of the resulting temporary decline in the Earth's temperature (about half a degree Celsius) were right on the money. He has been a force in convincing governments worldwide that global warming is something to be taken seriously.

How did Hansen get interested in the greenhouse effect in the first place? His doctoral thesis (at the University of Iowa in 1967) was about Venus. He agreed that the high radio brightness of Venus is due to a very hot surface, agreed that greenhouse gases keep the heat in, but proposed that heat from the interior rather than sunlight was the principal energy source. The *Pioneer 12* mission to Venus in 1978 dropped entry probes into the atmosphere; they showed directly that the ordinary greenhouse effect—the surface heated by the Sun and the heat retained by the blanket of air—was the operative cause. But it's Venus that got Hansen thinking about the greenhouse effect.

Radio astronomers, you note, find Venus to be an intense source of radio waves. Other explanations of the radio emission fail. You conclude that the surface must be ridiculously hot. You try to understand where the high temperatures come from and are led inexorably to one or another kind of greenhouse effect. Decades later you find that this training has prepared you to understand and help predict an unexpected threat to our global civilization. I know many other instances where scientists who first tried to puzzle out the atmospheres of other worlds are making important and highly practical discoveries about this one. The other planets are a superb training ground for students of the Earth. They require both breadth and depth of knowledge, and they challenge the imagination.

Those who are skeptical about carbon dioxide greenhouse warming might profitably note the massive greenhouse effect on Venus. No one proposes that Venus's greenhouse effect derives from imprudent Venusians who burned too much coal, drove fuel-inefficient autos, and cut down their forests. My point is different. The climatological history of our planetary neighbor, an otherwise